Big Data Analytics _

Telco Customer Churn Prediction and Intervention Prioritisation



Team members

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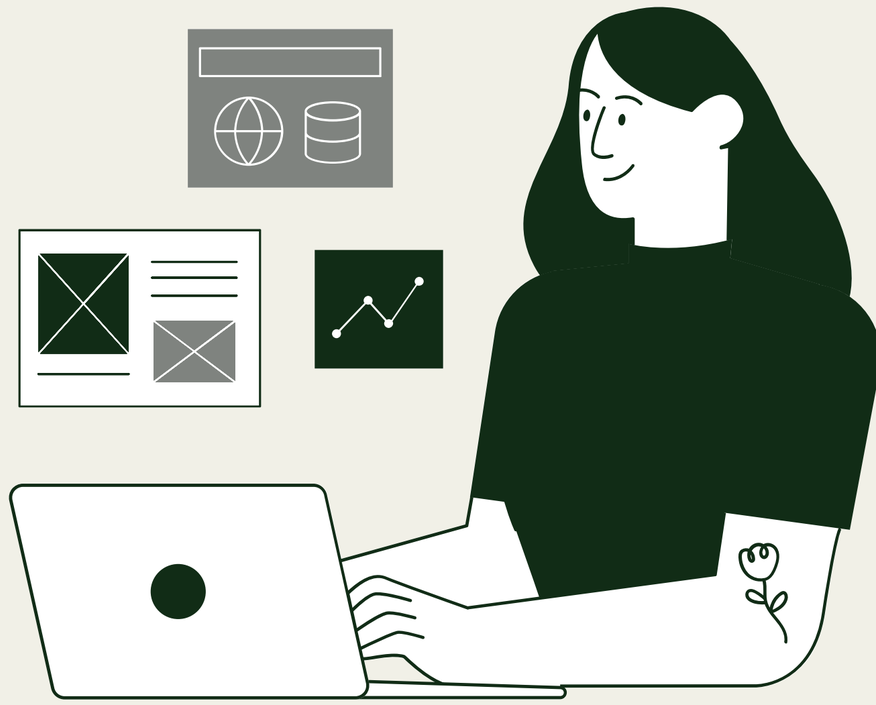


Table of contents:

01. The 4 Core Questions

02. Group SWOT Analysis

03. Team Communication Plan

04. Project Overview

05. Dataset & Method Selection

06. Proposed System Architecture

07. WBS for 4 weeks & Workspace setup

08. Research Paper Plan

4W Framework

1 What?

End-to-end Big Data pipeline (ML + Dashboard).
Week 1: Proposal, SWOT, Plan & Dataset.

2 When?

4-week roadmap (Preprocessing ➔ Modeling ➔ Final Report & Demo).

3 Why?

Practical application. Perfect scale for 3 members. Big Data/ML-ready dataset.

4 How much?

Planned: 5 hrs/week/pax. Actual (Week 1): 3 hrs total (Sourcing & scoping).

Group SWOT Analysis



Strengths

- Solid Python/SQL. Proven teamwork & synergy.
- Basic ML knowledge (Pandas, Scikit-learn).
- Strong problem-solving skills.



Weaknesses

- Limited Big Data (Spark/Hadoop) & Data Viz experience.
- Hardware constraints for local processing.



Opportunities

- Real-world practice. Master new tools (PySpark, PowerBI).
- Build a strong Portfolio/CV project.



Threats

- Noisy dataset (heavy cleaning).
- Schedule conflicts with other courses.
- Steep learning curve for new frameworks.

Team Communication



Channels

- Zalo (Daily/Urgent), Discord/Meet (Meetings/Pair-coding), Notion/Trello (Task management).



Schedule

Weekly: Thu, 8:00 PM (Discord). Ad-hoc: 4-hour notice.



SLA (Rules)

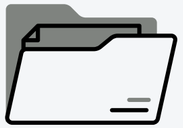
- Max 6-8h response. Pre-notify group before GitHub pushes.

Project Overview

Problems:

Telecommunications subscription model facing competitive pressure and rising customer acquisition costs.

Decision-maker:



Customer Retention Manager

Q1 – Driver Identification



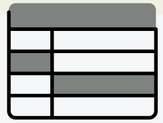
Which contract types, internet services, and payment methods are associated with the highest customer churn rates?

Q2 – Risk Prioritisation



Which customers have a predicted churn probability above 70% and should be prioritised for retention outreach?

Q3. Actionable Offer



Which retention incentives could help reduce churn while maintaining profit margins?

Dataset & Method Selection

SOURCES:

KAGGLE

(IBM TELCO CUSTOMER CHURN)

RECORDS:

7,043

(CUSTOMER RECORDS)

ATTRIBUTES:

21

(CUSTOMER ATTRIBUTES)

Feature and Target:

Group	Attributes
Demographics	Gender, Senior, Partner, Dependents
Services	PhoneService, InternetService, OnlineSecurity, TechSupport
Account	tenure, Contract, PaymentMethod, MonthlyCharges, TotalCharges
Target	Churn (binary: Yes / No)

Machine Learning Formulation:

Algorithm	Logistic Regression (binary: Churn = Yes / No)
Scoring	Confusion matrix, Precision, Recall, F1, ROC-AUC
Validation	80/20 stratified partitioning
Rationale	Sigmoid probabilities map directly to risk buckets for threshold tuning

Proposed System Architecture

1 Dataset ➔ MySQL

Kaggle Telco CSV (7,043 rows, 21 attributes) loaded into MySQL as telco_raw. SQL KPIs and filters run here.

2 KNIME ➔ ML

KNIME reads MySQL, fixes missing, duplicate and odd values, then trains Logistic Regression (80/20 stratified split).

3 Write-back to MySQL

Churn probability and risk tier per customer are saved to churn_predictions, the single source of truth for reporting.

4 Power BI ➔ Decisions

Power BI reads MySQL, flags the High risk tier (above 70% churn probability) and supports 3 retention actions.

WBS & Workspace Setup

Week 1 (M1): Repo and MySQL ready, proposal approved
Repo: github.com/[team]/telco-churn-retention. Effort: plan 15 hrs, actual 18 hrs (+3 hrs).

Week 2 (M2): SQL KPIs done, KNIME cleaning flow ready
Data lead: SQL KPIs and validation. Model lead: KNIME cleaning. Dashboard lead: wireframe.

Week 3 (M3): Predictions written to MySQL, dashboard v1
Model lead: Logistic Regression and write-back. Dashboard lead: three Power BI pages.

Week 4 (M4): Final report, GitHub repo and live demo
Whole team: 8–12 page report, README, demo rehearsal. Total effort: 5 hrs/week per member, 60 hrs.

Research Paper Plan

Working Title

Customer Churn Prediction and Retention Decision Support System Using KNIME and MySQL

Sections 1 to 3

- Introduction and decision questions
- Related work on churn prediction
- Data and methodology

Sections 4 to 6

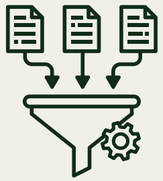
- Results: SQL KPIs, model, dashboard
- Decision support for Q1 to Q3
- Limitations and conclusion

Initial References

- Verbeke et al. (2012), EJOR
- Neslin et al. (2006), JMR
- Hosmer et al. (2013), Wiley
- Fawcett (2006), ROC analysis
- Provost and Fawcett (2013)

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Remove duplicates and errors



Handle missing values



Normalize and standardize data

Exploratory Data Analysis (EDA)

Key Steps:

Summarize data

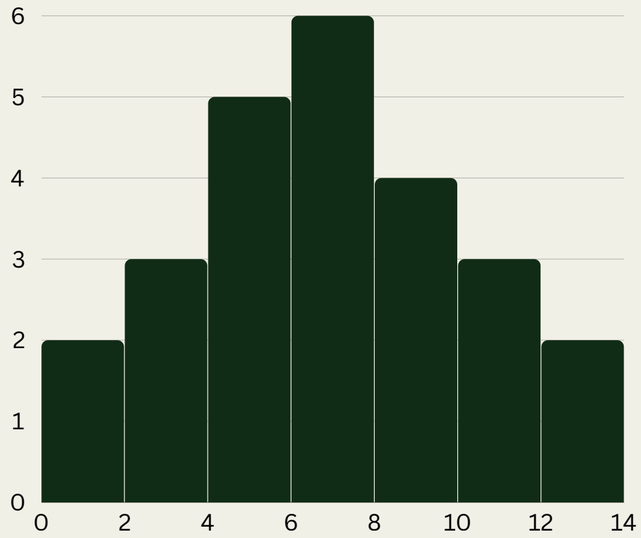


Visualize data



Identify patterns & correlations





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Case Study / Business Insights

Business Problem

Sales performance was inconsistent across regions and products

Data Used

- Sales transactions
- Product category
- Region & time period

Analysis Performed

- Trend analysis by month
- Comparison across regions and products

Analysis Performed

- Electronics showed the highest growth in Q2.
- Top 3 regions contributed over 60% of total revenue

Dashboard & Reporting



What is a Dashboard?

A dashboard is a visual tool that displays key metrics and performance in one view

Purpose of Reporting

- Monitor performance regularly
- Track key business metrics (KPI)
- Communicate insights to stakeholders

Outcome

Faster decision-making with clear and real-time insights

Challenges & Solutions

Challenges	Solution
Incomplete and inconsistent data across sources	Apply data cleaning and standardization process
Difficulty identifying meaningful patterns	Use data visualization and EDA techniques
Insights not easily understood by stakeholders	Present results through clear dashboards

Future Trends in Data Analytics

1 AI & Automation

Data analysis will be more automated using AI and machine learning

2 Real-Time Analytics

Businesses will rely on real-time data for faster decision-making

3 Advanced Data Visualization

Interactive dashboards and visual storytelling will become standard

4 Data-Driven Decision Making

Data will be used by all teams, not only analysts

Summary & Key Takeaways

Data Overview
The analysis was conducted using relevant data to identify key patterns, trends, and overall performance

Key Findings
Several important insights were identified that have a direct impact on performance and decision-making

Business Impact
The analysis highlights areas for improvement as well as opportunities for optimization

Action Points
The insights can be used as a foundation for defining the next strategic actions

Final Insights & Business Impact

Insight Generated

Data analysis reveals key patterns and opportunities

Business Impact

Insights support smarter and faster decisions

Value Delivered

mproved efficiency, performance, and strategy alignment



Closing Statement

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